

interest for analysis. Biomass-based binders can replace toxic halogenated commercial binders for holding the active material particles inside electrodes and ensure sustainability.

Numerous companies are adopting these technologies, and the situation is gradually changing. Last year, IBM developed a new type of battery that uses raw materials from seawater and

is free of cobalt, nickel and other heavy metals, thereby avoiding the trouble of mining and so reducing costs involved in the process. The battery, which is in an early stage of development currently, has a higher power density besides low flammability as compared to lithium-ion versions.

Sodium-ion is another option that has gained significant attention. Even

for lithium-sulphur batteries, it has been observed that the addition of molybdenum makes the electrodes conductive and stable. With further research and collaboration, it is possible to expect fully stabilised alternative materials that can be introduced in the market for real, practical, commercial applications in future.

—Ayushee Sharma

4 AI-DRIVEN COBOTS IN LOGISTICS AND BEYOND

With the advancements in technology, traditional robotic systems are now getting replaced by collaborative robots (cobots). Cobots are specifically designed to operate and simultaneously collaborate with human workers safely in the same workspace. These mainly cater to the manufacturing and logistics industries

One of the biggest changes in the Industry 4.0 revolution has been the use of robotics to automate mundane tasks. This was a big change for several organisations that boosted productivity by saving time as well as human labour. Once considered a novelty, robotics is now employed everywhere today, from offices, restaurants to even homes. With greater adoption over the course of time, robotics has evolved to incorporate changes necessary for its smooth functioning alongside humans.

One of the points stressed every now and then is the need to keep humans in the loop. This is why traditional robotic systems are getting replaced by collaborative robots (cobots). This is one of the fastest-growing segments of industrial automation.

Cobots are specifically designed to operate and simultaneously collaborate with human workers safely in the same workspace. For the cobots to be autonomous, data from sensors attached to them is fed to artificial intelligence (AI) models. This aids in collision avoidance and hence navigation as the exact position and velocity of the operators can be determined in real time. In case there is any failure in the system, the cobot automatically turns off. AI can even predict such conditions by drawing patterns from current and past data for better performance. There is no need for custom programming.

These robots are generally built to



AI-driven cobots (Credit: www.themanufacturer.com)

be lightweight and are easy to programme and reprogramme without the need for expertise in robotics. The payback period is also short. Most cobot manufacturers claim that the payback period lasts from about three to five months.

Besides AI, other emerging technologies like augmented reality (AR) can further help in accurate and precise real-time response. The bot that researchers at England-based Nottingham Trent University (NTU) have been working on for spinal surgery treatment uses AR to provide surgeons with live feedback.

The cobots mainly cater to the manufacturing and logistics industries. The reason is that there has always been a high demand for labour in the logistics sector ever since the rise of e-commerce, and amidst the lockdown restrictions during Covid-19 pandemic, this need is constantly growing. At a time when there are health risks involved in gathering a large number of workers at factories, cobots come to the rescue. Due to their

affordability and flexibility, even many small and medium-sized companies in India and abroad have deployed them for their business continuity.

Take the example of US-based Locus Robotics, which supplies its cobots to numerous logistics firms globally, such as CEVA Logistics and Geodis. The bots are capable of deploying sensors, carrying items between workstations, and direct human workers to the correct locations in any warehouse. Workers can focus on maximum picking, and the remaining work can be done by these machines.

In India too, cobots have high demand, especially in the automobile sector. Bajaj Auto has been using cobots for years, with more than 150 cobots currently across its assembly lines for purposes like machine tending.

With the world already grappling with unemployment, this poses a problem for many semi-skilled or unskilled workers who get daily wages for these routine tasks. These workers need to be upskilled before being given the newer job requirements that deal with preventive maintenance, validations, and the like.

While these bots have not found their place in homes yet, some companies have started making investments in this sector. In this nascent stage, Café X plans to begin shipping and installations of the beverage making equipment by the year-end, starting with the US. **EFY**

—Ayushee Sharma